

Natural Language Description

Refinery Process Production

```
- Flow\n\n```\nmermaid\ngraph LR\n  A1((Crude oil 1)) --> B[Distillation]\n  A2((Crude oil 2)) --> B\n  ...
```

Linear Modeling (Current methods and solver focus)

extract



$$A_1 + \sum_i b_i x_i \geq 0$$

Linear Math Model

Direct Mapping to Solver

SEMANTIC
Alignment

Solvers typically **cannot directly accept** such primitive nonlinear formulation.

```
...  
model.addConstr( A*x  
  <= b)  
...
```

Code Run
Success!

Nonlinear Modeling

extract



$$x^3 + \frac{1}{y} \leq c_i$$
$$C_1 \geq A_t e^{x_1 x_2^\beta}$$

Nonlinear Math Model

**SEMANTIC
GAP**

Coding Semantic

```
...  
model.addConstr(  
  A * x1**3 + 1/y <= C)  
...
```

⚠ Solver Cannot
Handle!